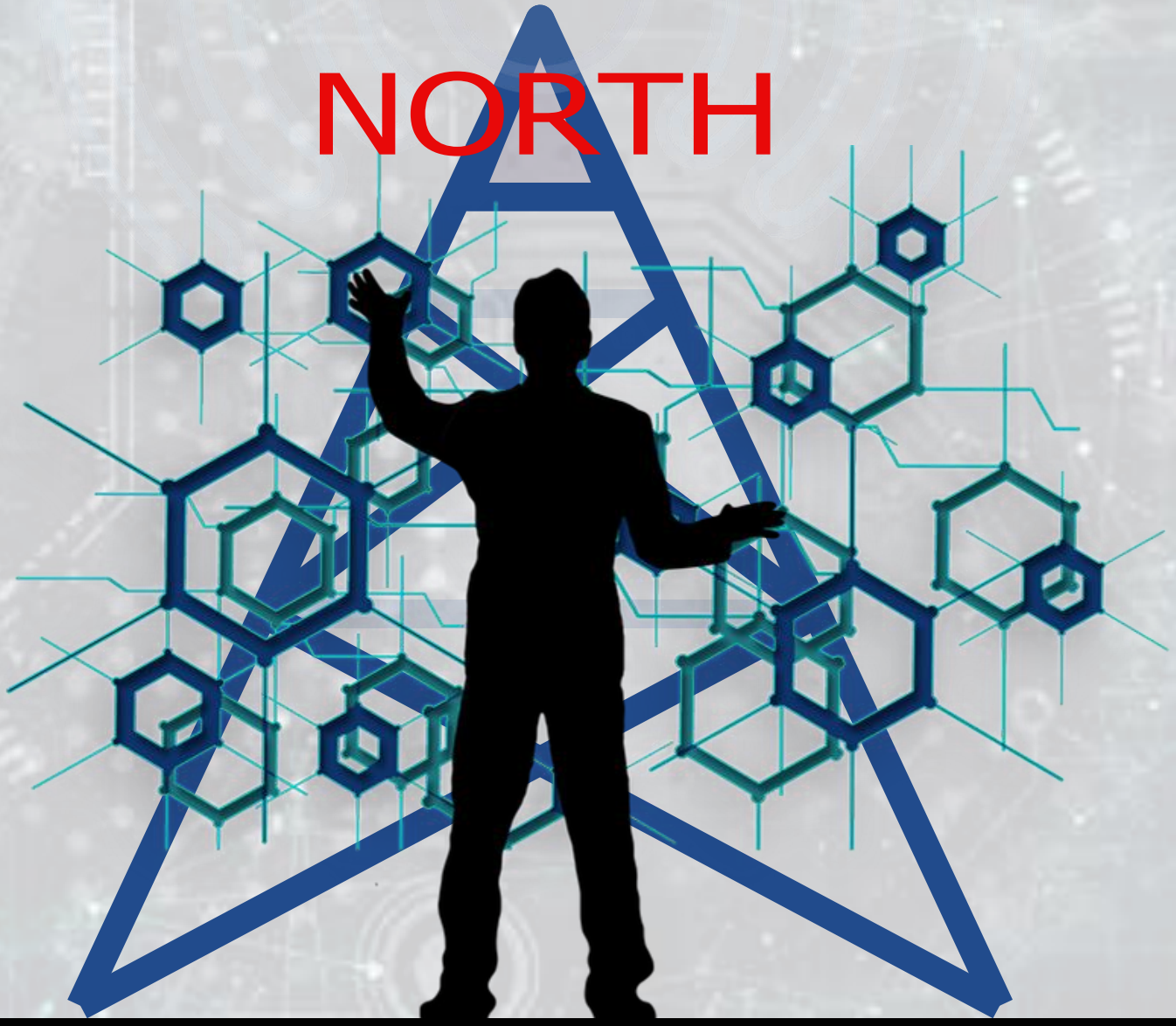
siberX

OPERATION DEFEND THE NORTH



ODTN Telecoms White Paper
Vancouver (April 2025)

By Gregory Onokwai



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1.0 EXECUTIVE SUMMARY

1.1. Brief Overview of Incident

A sophisticated cyberattack on April 16, 2025 triggered a total network blackout for ODTN Telecom Company. The incident commenced with customer complaints of disrupted wireless and internet services around the British Columbia region, Vancouver, soon followed by on-premise equipment in a network segment going offline. It quickly expanded to a total outage within a few hours. Investigations pinpointed an unauthorized, malicious firmware update delivered via a compromised management server, secured by weak SSH keys, as the root cause. This crippling attack exposed critical weaknesses in the company's operational infrastructure and its supply chain security.

1.2. Summary of the Defensive Operation

The defensive operation involved immediate network isolation and segmentation to contain the spread, disabling remote access, and terminating active sessions. A brand new "green" network was built, while the affected network had its nodes factory reset, reimaged, and reconfigured. Long-term strategies include enhanced password controls, MFA, and bastion hosts around privileged systems to bolster future resilience, and building a brand new environment.

1.3. Lessons Learned & Recommendations

- Proactive measures like continuous risk assessment, follow the right industry leading frameworks, cyber-awareness and user education training across the organization.
- More timely and efficient reactive measures like effective analysis, containment and eradication.



National Economic Impact: $ 46,760,249.00.
Total Time to Resolve Breach: 7h 15m 2s

- Data governance practices that would help if data privacy was at risk.
- Need to have tabletop exercises, practice internally & periodically for when crisis happens.
- Improved collaboration between SOC & NOC.
- Participate in sharing cycles of threat intelligence with peers and scan any IOCs learnt from that cycle on the network to see if they are present.
- Automate log monitoring process to reduce workload on SOC so they would have more time to carry out threat hunting.
- Correct inconsistencies around login, and asset inventory and management.
- Attack surface management and follow a defined playbook.
- Profile users not only compared against user history but against peers in the organization.

2.0 INTRODUCTION

2.1. Purpose of the White Paper

This white paper presents ODTN's (a fictional company) strong cybersecurity defense system against advanced attacks. It outlines our proactive approach, incident response strategy, and recovery mechanism that protect our Vancouver operations and key telecom infrastructure. The aim is to reassure stakeholders of our dedication to ensuring service integrity, data security, and operational resilience amidst emerging cyber threats.

2.2. Importance of Cybersecurity in Telecommunications Sector

Cybersecurity is vital in telecoms because it underpins the modern world. Telecoms networks transmit sensitive information, and as such, they are the major targets for cyber-attacks. Breaches result in financial loss, lawsuits, damage to reputation, and loss of customers' trust.



Availability of telecom services is vital for emergency services, critical infrastructure, and economic stability. DDoS attacks can disable these services, leading to severe disruption. Additionally, the introduction of 5G and IoT devices provides additional vulnerabilities. Strong cybersecurity is vital for commercial success, national security, and the integrity of consumers' confidence in communications service providers.



2.3. Overview of the Threat Landscape

The telecommunications industry is confronted with increasingly sophisticated and debilitating cybersecurity threats. Telecommunications providers are being attacked by numerous actors, ranging from state-sponsored actors to organized cybercriminals and even insiders. Common threats include sophisticated data breaches aimed at sensitive customer and operational data, Distributed Denial of Service (DDoS) attacks aimed at crippling critical services, and ransomware that can potentially cripple networks.

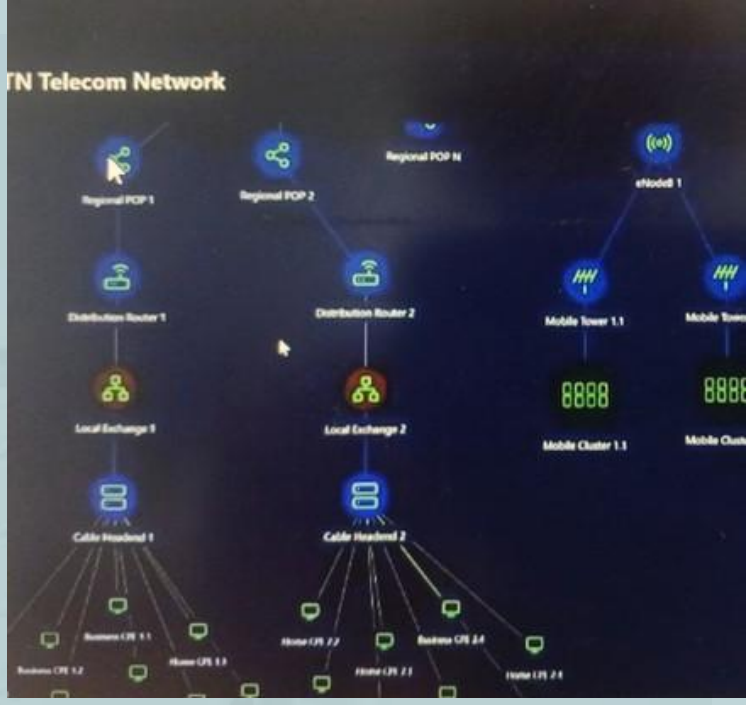
The proliferation of 5G and IoT brings in new vulnerabilities and a broader attack surface. Of particular interest are supply chain attacks, in which third-party vendors are compromised with the aim of infiltrating telecommunications networks. The motivations of such attacks are varied and include objectives such as financial gain, data exfiltration, espionage, and compromise of critical national infrastructure.

3.0 BACKGROUND

3.1. Network Architecture Overview

To maximize efficiency and further allow scalability options, the network is organized at different levels hierarchically. The Core Network serves as the backbone that connects data centers and the internet and manages heavy traffic.

The Layer beneath it aggregates traffic from access points and sends it to the core, merging service delivery and security functions. The Access Layer gives end-users (businesses and homes) access to devices like mobile phones and broadband. It ensures dependable and efficient delivery of services.



3.2 Security Posture Prior to Incident

- Internetwork was up and running.
- Baselines were normal.
- Alerts received were normal.
- No report of any breaches.
- Data was flowing.

3.3 Regulatory & Compliance Environment

Adherence to cybersecurity in Canada involves following regulations and standards by bodies like the Canadian Centre for Cybersecurity (CCC), Canadian Radio-Television and Telecommunications Commission (CRTC), Canadian Cyber-Threat Exchange (CCTX), Royal Canadian Mounted Police (RCMP), and Public Safety of Canada in order to protect digital information and infrastructure. Organizations must prioritize cybersecurity compliance to protect sensitive data and maintain stakeholder trust. Best practices can mitigate risks from cyber threats.

4.0 MODULES

1 DETECTION & ANALYSIS

4.1.1. Initial IOCs

- The earliest was when the media word came from a few customers of disruption of their wireless & internet services.
- On premise equipment (Business & Home) went offline on the dashboard.

4.1.2. Scope & Origin of Outage

- Scope: Initially British Columbia Area but later became nationwide from government advisory.
- Origin: Advanced Persistent Threat (APT) actors similar to Salt Typhoon

4.1.3. Investigation & Communication

Strategies used were

- Zero Trust
- Root Cause Analysis
- Internal Investigation
- Crisis Containment

Strategies used for Communication

- Out-of-Band communication
- Privileged Communication

2 CONTAINMENT

4.2.1 Key systems and services stabilized during outage by

- Isolation & logical segmentation of affected systems.
- Put into action BCP and IRP, and ensured continuum of services of critical infrastructure.
- Reach out to trusted partners, other telecoms

for support.

- Established out of band communication.
- Disable remote access

4.2.2 To isolate affected components, secure alternative communication methods, and prevent further disruption to essential operations,

- Silos were kept in place to segment the network
- Disabled all remote access to the affected systems.
- Killing off all active sessions.
- VLAN isolation & segmentation of compromised nodes
- Invoke privileged communication.
- Used out-of-band communication.

3 ERADICATION

4.3.1 The root causes of the outage identified and addressed via

- The NOC & SOC teams found a change in the log that was pushed across the network through a management server (the internet update server) while perusing the logs.
- The firmware and operating system were fixed and configuration recovered.

4.3.2 The steps that were taken to eliminate threats, restore functionality, and limit the impact on interconnected systems are

- The patch for the firmware that caused the issue was deployed.
- Factory reset the affected nodes, reimaged and reconfigured them.
- .Performed rootkit scans to confirm there were no persistent malwares still present in the system.
- Appl ACL to limit lateral movement to stop spread.

4 RECOVERY

4.4.1 Operations gradually restored as systems came back online via

- Clean configurations, also made sure the new systems are in place.
- Validated all kernel modules.
- Made sure checksums are in place.
- Validate system health.
- Firmware and OS working properly

4.4.2 The measures implemented to ensure stability, verify system integrity, and prioritize the resumption of essential services are

- Accounts were flipped, reset
- Only approved individuals were allowed to have access to the devices.
- Effective monitoring.
- Broad password resets

5 POST INCIDENCE ACTIVITY

4.5.1 Long-term strategies to strengthen resilience and prevent future incidents were

- Practices around continuous risk assessment and tabletop exercises.
- A need to follow industry leading frameworks.
- Data governance practices & cyber-awareness training.
- Jump boxes (bastion) around
- privileged systems.
- One time password from a password
- tool like cyberArk on high privileged systems

4.5.2 Public trust was restored, and lessons carried forward to improve national response efforts in future large-scale cyber disruptions were

- A new international resilience standard for telecom
- The need for countries to share intelligence, resources and response protocols in cross border cooperation

5.0 RELATE MODULES TO CYBERSECURITY PRINCIPLES

5.1. Zero Trust Architecture: This would have prevented lateral movement within the network by

- Eradicating implicit trust and verifying all access requests.
- Micro-segmentation of the network into isolated segments would have prevented the attacker from moving to other segments without re-authentication.
- Applying least privilege access which would have restricted the attacker's activities despite the compromised credentials.
- Continually verifying and authenticating user identity, device health, location, and other factors which would have severely mitigated the attack.

5.2. Supply Chain Security: Dependencies on third-party infrastructure revealed vulnerabilities in national communication systems through

- Unpatched vulnerabilities, backdoors or vulnerable codes by equipment manufacturers allowed attacks and provided the attacker with access to the network.
- The firmware installed was not from the vendor. A rogue version of the firmware was put in the environment.
- Poor configurations.

5.3. Incident Response & Crisis Management

- They had IRP, BCP & DRP
- They set up two war rooms: Command and Control Structure: One technical room with an incident commander and another for C level executives.
- They deployed distributed incident response teams (IRT) to different geographic locations and had established out-of-band communication and off-line procedures for that kind of situation.
- They could access funds from their insurer for a new, green network, had some redundant and secure equipment that could be deployed and leveraged on resources of and MOU agreements with their partners.

5.4. Data Privacy & Protection: What safeguards protected sensitive records during the outage in sectors like healthcare and finance?

- Used offline and local access control policies (e.g., offline MFA, hardware tokens).
- Stored critical records in physically and logically segmented systems that do not rely on real-time telecom connectivity.
- Offline backups in secure, access-controlled locations and enabled manual record-keeping protocols (e.g., paper-based patient logs or financial transaction ledgers).
- Regulatory Compliance and Legal Safeguards: Like HIPAA (Healthcare) – ensure PHI confidentiality under 45 CFR §164.306 during contingencies and GLBA or PCI DSS (Finance) – maintain physical and logical access controls, even offline.

5.5. Cyber Resilience: How soon and successful an organization's recovery goes depends on how well prepared it was before attack, structures it





had in place and how responsive and strategic they were. Structures needed include

- IRP + BCP to guide organized, prioritized response.
- Crisis Command Team to provide leadership and decision-making clarity.
- Segmentation of network & backups of data to minimize data loss and lateral spread.
- Tested communication plan to maintain trust with staff, customers, regulators.
- Postmortem and rebuilding to improve defenses and avoid recurrence.

6.0 RELATE TO REAL-WORLD INCIDENTS AND QUOTES

Module 1 – Detection & Analysis:

The outage at Optus was first identified at approximately 4:05 AM AEDT on November 8, 2023. Customers began reporting service disruptions, and Optus's internal monitoring systems detected anomalies in network performance. The initial communication was delayed and sparse. The outage began at 4.05 AM AEDT, the initial response was several hours later at 9.00am which led to customers frustration.

Optus communicated primarily via X (formerly Twitter), Facebook, and their website which was inaccessible to many due to the outage affecting mobile data and broadband especially those who rely solely on Optus for internet access. They provided hourly updates, though many were vague and offered no estimated time for resolution. Optus lacked a pre-defined strategy for communicating during a blackout such as analog radio, direct government channels, or press briefings.

"The CEO may be the CEO but the CEO is still beholden to the law in Canada" Richard Henderson



"In Incidence Response, currency is time."
Geoff Rivers

Module 2 – Containment:

The 2021 Colonial Pipeline ransomware attack demanded immediate containment efforts to prevent widespread infrastructure disruption and economic fallout.

- Colonial proactively shut down operations across its entire pipeline system shortly after detecting the ransomware on its IT network to prevent spread and ensure integrity. This was a preventive containment measure to isolate operational networks and prevent potential ransomware movement into OT environments.
- Colonial had some degree of segmentation between IT (business systems) and OT (pipeline operations). Likely prevented direct compromise of physical pipeline control systems.
- Colonial notified the FBI, DHS, and CISA early in the response and brought in Mandiant (now part of Google Cloud) to perform forensics, containment, and remediation.
- They isolated infected IT systems, began root cause analysis, and advised on restoration strategies. Colonial paid the $4.4 million ransom to obtain a decryption tool from the attackers.

Module 3 – Eradication:

"They say that the best time to relax is when you have no time for it"
Ali Abbas Hirji

- The June, 2017 NotPetya malware Outbreak which affected multiple companies across several countries
- Ukraine was hardest hit, the irreversible nature of the malware, led to the quick shut down all systems to stop its spread of the malware. Systems could not be repaired; they had to be rebuilt from scratch.
- Microsoft, Cisco, and other vendors were brought in to support a 24/7 rebuild operation.
- A global IT reconstruction effort across over 600 sites in 130 countries using a single surviving domain controller in a remote office in Ghana. With Ghana DC and clean backups, Maersk re-imaged thousands of systems in a matter of days.
- Operational challenges include global downtime, communication loss, shipping disruptions, reputation risk, and resource bottlenecks.



Module 4– Recovery:

On July 8, 2022, a configuration error during a maintenance update led to the removal of a critical Access Control List (ACL) filter from distribution routers in Rogers' IP network. In a nationwide service outage that lasted over 15 hours. Rogers,

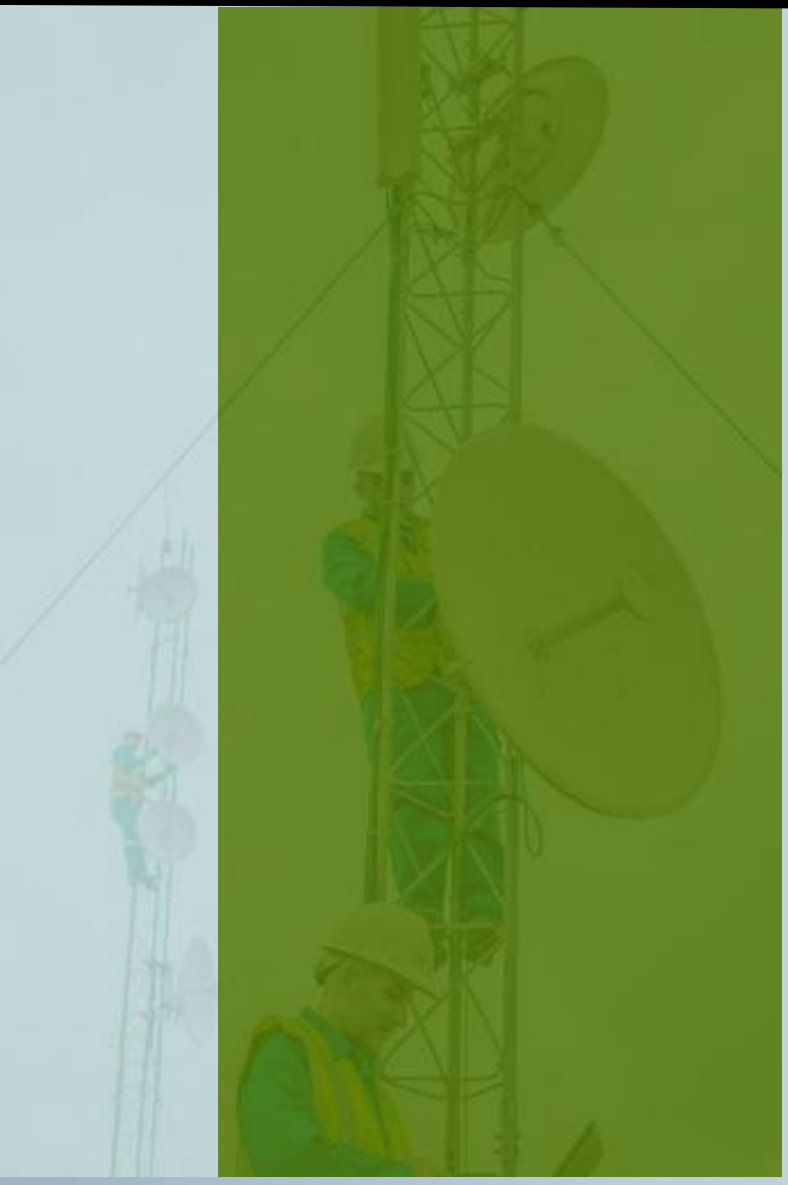
- Prioritized the restoration of wireless and wireline services, followed by critical care services and major business customers. The company initiated a phased recovery process, gradually bringing services back online to prevent further overload and ensure system stability.



- Enhanced network resilience, and implemented safeguards and redundancy
- Introduced a new risk assessment algorithm, improved collaboration between network operations and engineering teams, and implemented enhanced process for introducing new equipment and technology.
- Publicly apologized and compensated customers by offering service credits and rebates, estimated between $28 million to $70 million, to affected users.
- Kept customers informed during the restoration process using social media, media outlets and customer service platforms.
- Signed a Memorandum of Understanding (MoU) with other major Canadian telecom providers, including Bell and Telus, aimed at ensuring mutual assistance during future network outages.



"NEVER WASTE A GOOD CRISIS"
NEUMAN LIM



Module 5 – Post-Incident Activity:

The 2020 SolarWind supply chain attack involved the compromise of its Orion software, which was pushed as a legitimate update to over 18,000 customers.

- This led to agencies and enterprises scrutinizing their software supply chains and Increased requirements for code integrity, third-party audits, and software provenance.
- Push toward continuous monitoring of software behavior post-deployment.
- Promoted a "no-fault" culture of early disclosure and shared intelligence

> **"Compliance does not equal security"**
> **Octavia Howell**



7.0 REFLECTING FINDINGS ACROSS CYBERSECURITY DISCIPLINES

7.1. Telecom Infrastructure

- To avoid outages and reduce single point of failure, provider and geographical redundancy are required by Telecom networks.
- Edge computing and local data centers should power core services while dynamic rerouting of failures should be supported by software-defined networking (SDN).
- Agreements to route traffic on another telecom company's infrastructure in times of network failure.

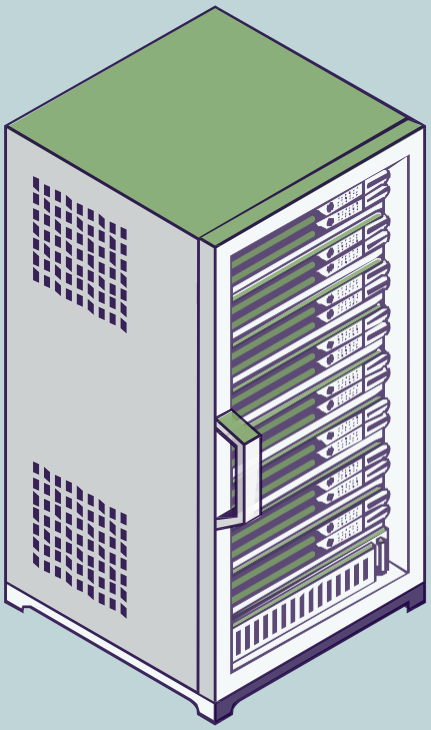
7.2. Law and Policy:

- Compulsory reporting of outages by telecoms and response times should be required by governments and mandatory data sharing among telecoms and public authorities during a crisis.
- A National Telecom Resilience Act can require redundancy, failover systems, multi-provider support and should include mandatory compliance auditing, penalties for negligence, and uptime standards.
- Finally, the legislation needs to establish independent governance to oversee national telecom resilience.



7.3. Public Sector Preparedness:

- Clear communication, collaboration and leadership should be the basis of effective emergency management by federal, provincial, and municipal agencies.
- A unified national framework that defines roles, responsibilities, escalation procedures, and resource allocation during a crisis.
- Joint training exercises across all levels of government to ensure seamless interaction and readiness.
- Standardizing and securing the sharing of real-time data and outage information.



7.4. Cybersecurity Frameworks:

- Maersk initially showed obvious gaps in adhering to the NIST and ISO 27001 "Protect" principles, particularly regarding patch management and network segmentation which are fundamental components of ISO-aligned Information Security Management System.

- Maersk's efforts to reconstruct their infrastructure closely reflected the NIST "Recover" phase.

- Their transparent communication post-incident embodied the information-sharing objectives outlined in CISA's National Cyber Incident Response Plan.

8.0 Reactive Incidence Response Timeline

Time Since Breach	IR Stage	IOC	Source	Actions Taken
9.00am	Detection & Analysis	• Reports of a supposed breach	• SiberX News • Word from Customers	• Investigating the reports. • Internetwork is up and running. • No signs of a breach.
0h 46m 28s	Detection & Analysis	On Premise Home & Business Equipment went offline	• SIEM Tool • SOAR	• Devices going offline happens at times. • Possible breach unconfirmed
1h 49m 8s - 2h 37m 4s	Containment	• Multiple devices went offline. • Christmas tree lighting up.	• SIEM tool • SOAR	• Out of band communication. • Apply ACLs • Disable remote access. • VLAN Isolation and logical segmentation of affected systems. • Killing active sessions.
3h 42m 19s - 4h 42m 1s	Eradication	• Action the IRP & BCP, and jump boxes & flash firmware Activate • zero trust and create 2 war rooms: Technical & C-level. Action • sanitization instructions from third party experts and clean up the system.		
5h 17m 45s - 6h 17m 15s	Recovery	• Build a new green network deploying redundant equipment in the warehouse and integrate it to existing clean environment. • Make mandatory report to CRTC about recovery. • Accounts flipped and reset, and access controls in place.		
6h 45m 13s - 7h 15m 1s	Post Incident Activity	Reflecting on the incidence, suggestions were made on way forward: • Participation in sharing cycles of threat intelligence from peers. • Removing inconsistencies around user login, MFA, asset inventory & Management and profiling users. • Having basic visibility across your digital footprint.		

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